

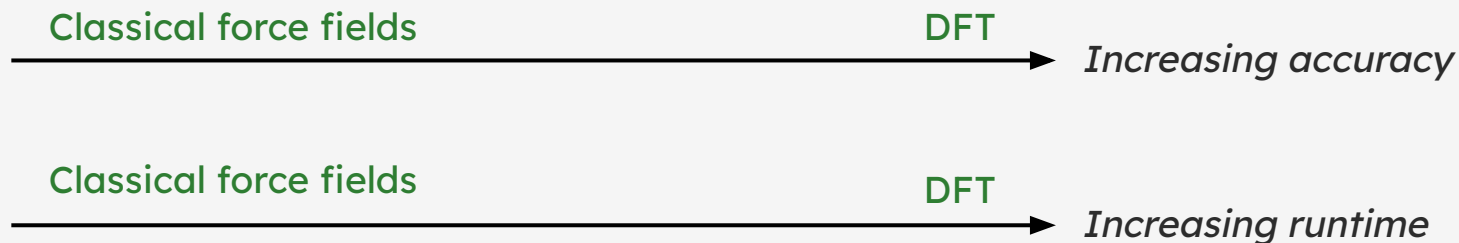
Implementing an AMBER force field in C++

CHEM 279 Final Project

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Intro to force fields

- A model that describes the forces between atoms
- 3 broad classes
 - Class 1: bond stretching and angle bending described by harmonic motion (e.g., AMBER) → today's presentation
 - Class 2: anharmonic terms; cross-coupling between adjacent bonds/angles; (e.g., MMFF94)
 - Class 3: specific chemical terms such stereoelectronic effects, polarization, etc. (e.g., AMOEBA)



Functional form of the AMBER force field

$$E_{\text{total}} = \sum_{\text{bonds}} K_r (r - r_{\text{eq}})^2 + \sum_{\text{angles}} K_{\theta} (\theta - \theta_{\text{eq}})^2 + \sum_{\text{dihedrals}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^6} + \frac{q_i q_j}{\epsilon R_{ij}} \right]$$

Term 1 Term 2 Term 3 Term 4

K_r : bond spring constant

r_{eq} : equilibrium bond length

K_{θ} : angular spring constant

θ_{eq} : equilibrium bond angle

V_n : barrier height

n : periodicity

ϕ : dihedral angle

γ : phase shift

A_{ij} : repulsion coefficient

B_{ij} : attraction coefficient

Term 1: bond stretching (harmonic approximation)

Term 2: angle bending (harmonic approximation)

Term 3: torsion (rotation around bonds)

Term 4: nonbonded interactions (van der Waals, electrostatics)

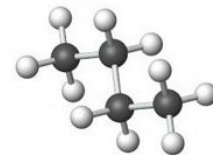
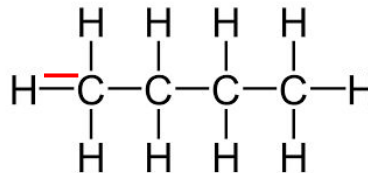
Codebase overview: force field parameters

- Suppose our molecule is a linear alkane
- 2 atom types in the molecule:
 - sp³ carbon
 - Hydrogen
- Obtain comprehensive list of parameters from [ambermini package](#) and extract those we need

```
# Bond term: E = K_r * (r - r_eq)^2
# format: type_i type_j K_r [kcal/mol/angstrom^2] r_eq [angstrom]
[bonds]
c3 c3 232.52 1.538
c3 hc 375.92 1.097
```

Bond Stretching Energy

BUTANE STRUCTURE



Loops over all direct bonds

$$\sum_{\text{bonds}} K_r (r - r_{eq})^2$$

Term 1

K_r : bond spring constant
 r_{eq} : equilibrium bond length

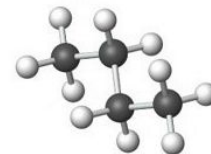
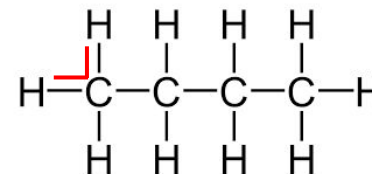
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```
# Angle term: E = K_theta * (theta - theta_eq)^2
# format: type_i  type_j  type_k  K_theta [kcal/mol/rad^2]  theta_eq [deg]
[angles]
c3 c3 c3    64.888  111.510
c3 c3 hc    46.816  109.800
hc c3 hc    38.960  107.580
```

Angle Bending Energy

BUTANE STRUCTURE



Loops over all pairs of neighboring bonds

$$\sum_{\text{angles}} K_{\theta} (\theta - \theta_{\text{eq}})^2$$

Term 2

K_{θ} : angular spring constant
 θ_{eq} : equilibrium bond angle

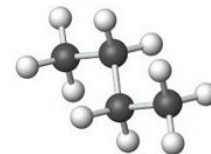
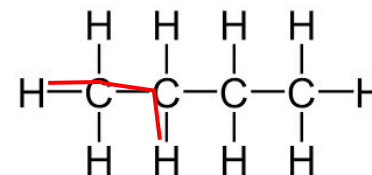
Codebase overview: force field parameters

- Suppose our molecule is a linear alkane
- 2 atom types in the molecule:
 - sp³ carbon
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```
# Dihedral term: E = (Vn / divider) * (1 + cos(n*phi - gamma))
# format: type_i type_j type_k type_l divider Vn gamma [deg] n
[dihedrals]
c3 c3 c3 c3 1 0.130 0.000 3
c3 c3 c3 c3 1 0.290 180.000 2
c3 c3 c3 c3 1 0.110 0.000 1
c3 c3 c3 hc 1 0.080 0.000 3
hc c3 c3 hc 1 0.120 0.000 3
```

Torsion Energy

BUTANE STRUCTURE



$$\sum_{\text{dihedrals}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)]$$

Term 3

V_n : barrier height
 n : periodicity
 γ : phase shift

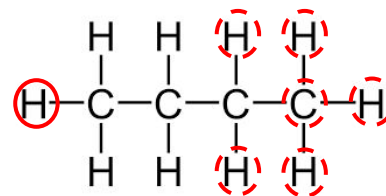
Codebase overview: force field parameters

- Suppose our molecule is a linear alkane
- 2 atom types in the molecule:
 - sp³ carbon
 - Hydrogen
- Obtain comprehensive list of parameters from [ambermini package](#) and extract those we need

```
# vdW term: E = eps_ij * [(R*_ij/r)^12 - 2*(R*_ij/r)^6]
# format: type R_star [angstrom] epsilon [kcal/mol]
[vdw]
c3  1.9069  0.1078
hc  1.4593  0.0208
```

Van der Waals Energy

BUTANE STRUCTURE

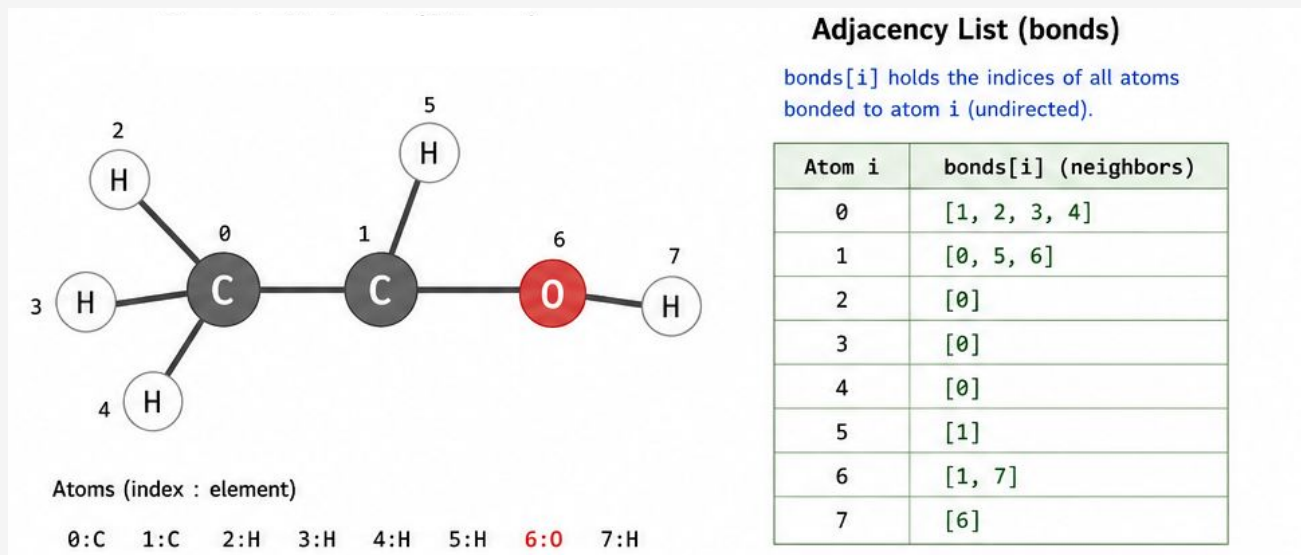


$$\sum_{i < j} \left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^6} \right]$$

Term 4

A_{ij} : repulsion coefficient
 B_{ij} : attraction coefficient

Codebase overview: molecule as a graph

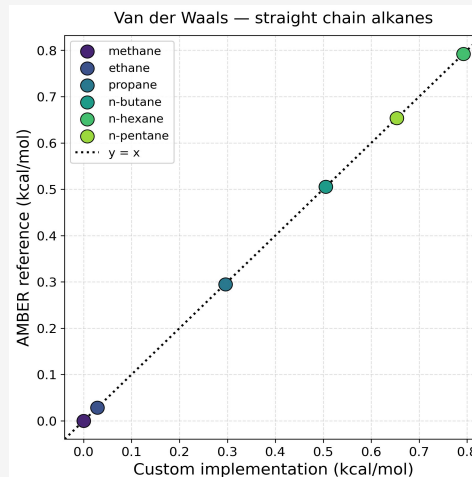
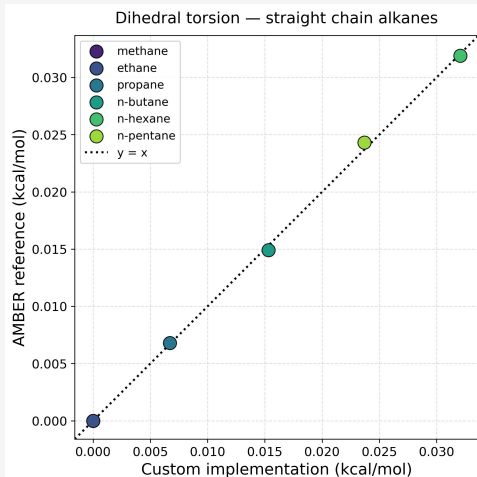
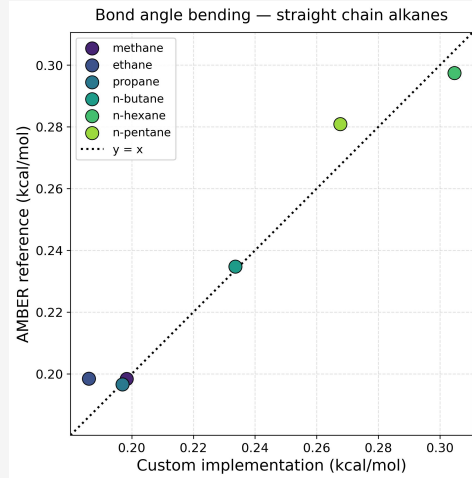
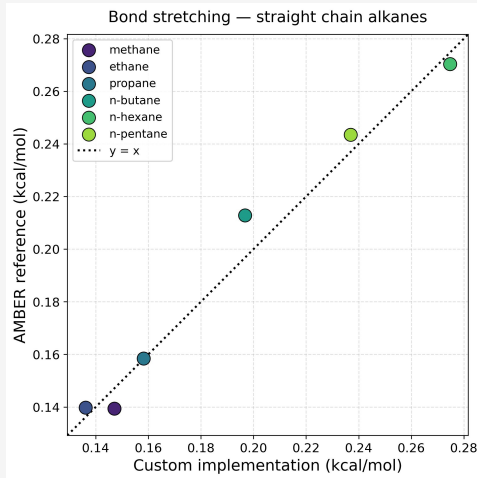


- Only atom coordinates as input
- Bonds inferred by interatomic distances
- Atom's “AMBER type” assigned - covering alkanes through amino acids

Keeps track of bonds and atoms, enabling us to identify all bond angles (triplets) and all torsion (quadruplets)

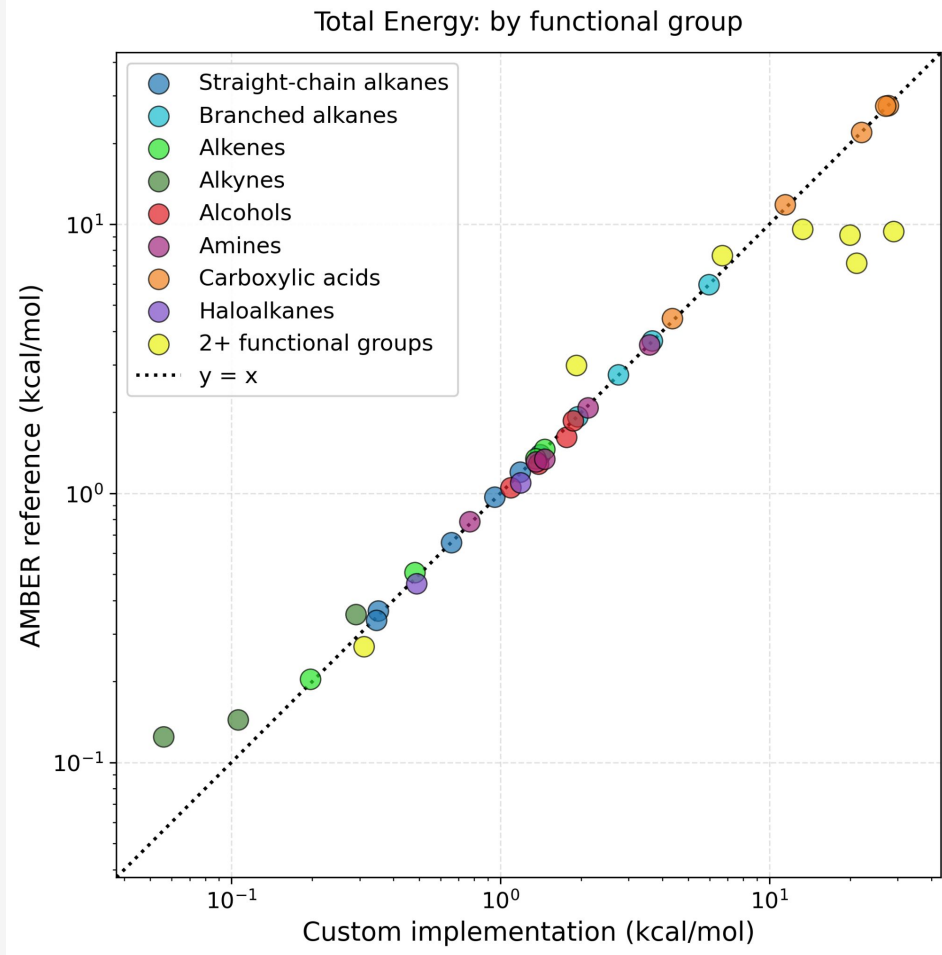
Benchmarking study: straight chain alkanes

- Benchmarked against the actual AmberTools25 pipeline
- All four energy terms track the benchmark closely
- Energies increase (almost linearly) with chain length



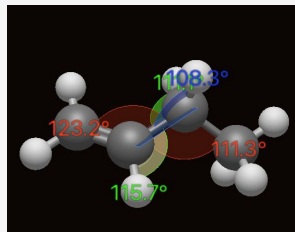
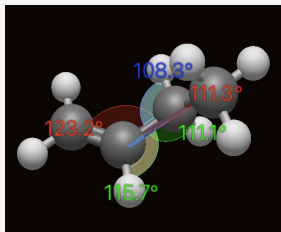
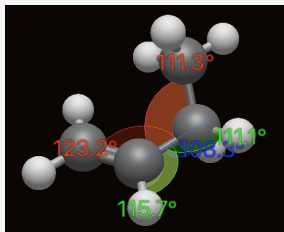
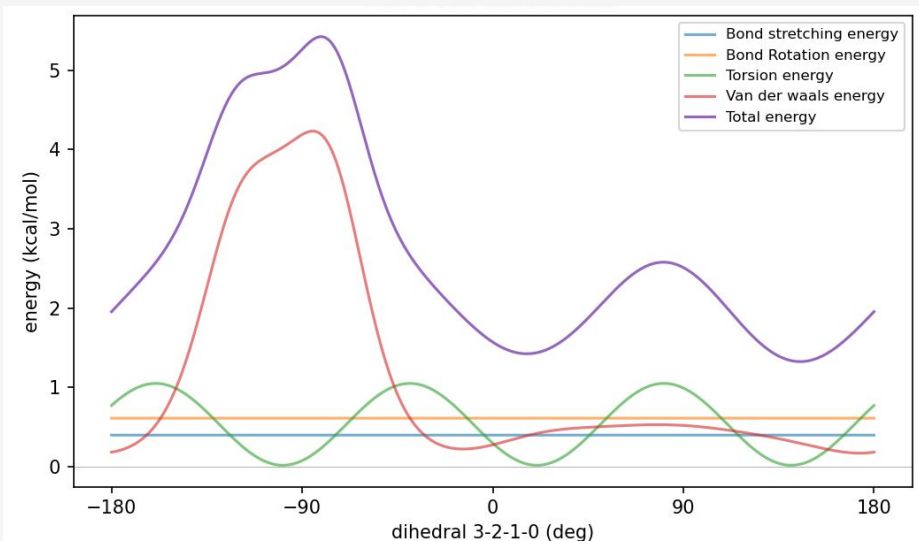
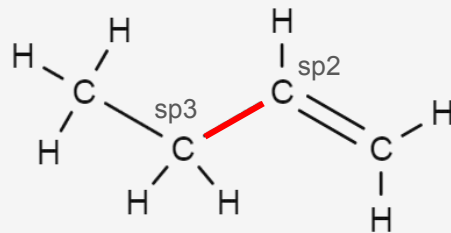
Benchmarking study: All functional groups

- Clustering by group:
 - Alkynes are rigid (low torsion energies)
 - Carb. acids, amino acids, etc have more heavy atoms (O,N,Cl) and more degrees of freedom
- Incorrect typing causes slight deviations from benchmarks
 - errors are largest on mixed group/conjugated or polar molecules where specialized atom-types matter most, and smallest on pure hydrocarbons



Torsion Scan: Conformers of but-1-ene

- Vdw energy is max when the -CH₃ on C(sp³) is close to the -H on sp² C(sp²)
- Torsion energy oscillates every 120 deg
- Bond stretching and angle bending energies are basically constant



Takeaways and significance of the AMBER force field

- Force fields bridge the gap between QM properties and macroscopic observables, by allowing us to scale to systems with millions of atoms
- Only 5 analytical terms (including Coulombic interactions) used for AMBER. Complexity comes from the extensive parameter list
- Reading parameters based on atom types is crucial for accuracy, but that limits its scope. Cannot capture bonds breaking or polarization effects